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Part 1: Test requirements and Protocol Implementation Conformance Statement (PICS) pro forma

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Contents

Intellectual Property Rights 4
Foreword 4
Modal verbs terminology 4
1 Scope 5
2 References 5
2.1 Normative references 5
2.2 Informative references 5
3 Definition of terms, symbols and abbreviations 5
3.1 Terms 5
3.2 Symbols 6
3.3 Abbreviations 6
4 Conformance requirement concerning PICS 6
Annex A (normative): IPV6overGEONETW PICS Pro forma 7
A.1 The right to copy 7
A.2 Guidance for completing the ICS pro forma 7
A.2.1 Purposes and structure 7
A.2.2 Abbreviations and conventions 7
A.2.3 Instructions for completing the ICS pro forma 9
A.3 Identification of the implementation 9
A.3.1 Introduction 9
A.3.2 Date of the statement 9
A.3.3 Implementation Under Test (IUT) identification 9
A.3.4 System Under Test (SUT) identification 10
A.3.5 Product supplier 10
A.3.6 Client (if different from product supplier) 11
A.3.7 ICS contact person 11
A.4 Identification of the protocol 12
A.5 Global statement of conformance 12
A.6 Tables 12
A.6.1 Introduction 12
A.6.2 ITS Station type 12
A.6.2.1 General 12
A.6.2.2 IPv6OverGeonetW sub layer 12
A.6.2.3 Router type 12
A.6.3 Virtual links 13
A.6.4 Virtual network interfaces 13
A.6.5 Primitives Associated 13
A.6.6 Packet Delivery 13
A.6.7 IPv6 14
A.6.7.1 IPv6 cast 14
A.6.7.2 IPv6 multicast 14
A.6.8 IPv6 neighbour discovery support 14
A.6.8.1 IPv6 neighbour discovery 14
A.6.8.2 Address configuration 14
A.6.8.3 Address resolution 14
A.6.8.4 Constants 15
A.6.9 Protocol parameters 15
History 16

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Foreword

This Technical Specification (TS) has been produced by ETSI Technical Committee Intelligent Transport Systems (ITS).

The present document is part 1 of a multi-part deliverable covering Conformance test specifications for Transmission of IP packets over GeoNetworking, as identified below:

Part 1: "Test requirements and Protocol Implementation Conformance Statement (PICS) pro forma";

Part 2: "Test Suite Structure and Test Purposes (TSS & TP)";

Part 3: "Abstract Test Suite (ATS) and Protocol Implementation eXtra Information for Testing (PIXIT)".

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "will not", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

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1 Scope

The present document provides the Protocol Implementation Conformance Statement (PICS) pro forma for Conformance test specifications for Geonetworking Basic Transport Protocol as defined in ETSI EN 302 636-6-1 [1] in compliance with the relevant requirements and in accordance with the relevant guidance given in ISO/IEC 9646-7 [i.2].

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

Referenced documents which are not found to be publicly available in the expected location might be found at https://docbox.etsi.org/Reference.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long term validity.

The following referenced documents are necessary for the application of the present document.

[1] ETSI EN 302 636-6-1 (V1.2.1): "Intelligent Transport Systems (ITS); Vehicular Communications; GeoNetworking; Part 6: Internet Integration; Sub-part 1: Transmission of IPv6 Packets over GeoNetworking Protocols".

[2] IEEE 802.3™-2012: "IEEE Standard for Ethernet".

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long term validity.

The following referenced documents are not necessary for the application of the present document but they assist the user with regard to a particular subject area.

[i.1] ISO/IEC 9646-1 (1994): "Information technology - Open Systems Interconnection - Conformance testing methodology and framework - Part 1: General concepts".

[i.2] ISO/IEC 9646-7 (1995): "Information technology - Open Systems Interconnection - Conformance testing methodology and framework - Part 7: Implementation Conformance Statements".

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in ETSI EN 302 636-6-1 [1], ISO/IEC 9646-1 [i.1] and in ISO/IEC 9646-7 [i.2] apply.

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

DGVL	Dynamic Geographical Virtual Link
ICS	Implementation Conformance Statement
IP	Internet Protocol
IPv6	Internet Protocol version 6
ITS	Intelligent Transportation Systems
IUT	Implementation Under Test
LAN	Local Area Network
PDU	Protocol Data Unit
PICS	Protocol Implementation Conformance Statement
SGVL	Static Geographical Virtual Link
SUT	System Under Test
TVL	Topological Virtual Link

4 Conformance requirement concerning PICS

If it claims to conform to the present document, the actual PICS pro forma to be filled in by a supplier shall be technically equivalent to the text of the PICS pro forma given in annex A, and shall preserve the numbering, naming and ordering of the pro forma items.

An ICS which conforms to the present document shall be a conforming PICS pro forma completed in accordance with the instructions for completion given in clause A.1.

Annex A (normative): IPV6overGEONETW PICS Pro forma

A.1 The right to copy

Notwithstanding the provisions of the copyright clause related to the text of the present document, ETSI grants that users of the present document may freely reproduce the IPV6overGEONETW PICS pro forma in this annex so that it can be used for its intended purposes and may further publish the completed IPV6overGEONETW PICS.

A.2 Guidance for completing the ICS pro forma

A.2.1 Purposes and structure

The purpose of this PICS pro forma is to provide a mechanism whereby a supplier of an implementation of the requirements defined in ETSI EN 302 636-6-1 may provide information about the implementation in a standardized manner.

The PICS pro forma is subdivided into clauses for the following categories of information:

- guidance for completing the ICS pro forma;
- identification of the implementation;
- identification of the ETSI EN 302 636-6-1;
- global statement of conformance;
- PICS pro forma tables.

A.2.2 Abbreviations and conventions

The ICS pro forma contained in this annex is comprised of information in tabular form in accordance with the guidelines presented in ISO/IEC 9646-7.

Item column

The item column contains a number which identifies the item in the table.

Item description column

The item description column describes in free text each respective item (e.g. parameters, timers, etc.). It implicitly means "is <item description> supported by the implementation?".

Status column

The following notations, defined in ISO/IEC 9646-7, are used for the status column:

m	mandatory - the capability is required to be supported.
o	optional - the capability may be supported or not.
n/a	not applicable - in the given context, it is impossible to use the capability.
x	prohibited (excluded) - there is a requirement not to use this capability in the given context.
o.i	qualified optional - for mutually exclusive or selectable options from a set. "i" is an integer which identifies an unique group of related optional items and the logic of their selection which is defined immediately following the table.
ci	conditional - the requirement on the capability ("m", "o", "x" or "n/a") depends on the support of other optional or conditional items. "i" is an integer identifying an unique conditional status expression which is defined immediately following the table.
i	irrelevant (out-of-scope) - capability outside the scope of the reference specification. No answer is requested from the supplier.

NOTE 1: This use of "i" status is not to be confused with the suffix "i" to the "o" and "c" statuses above.

Reference column

The reference column makes reference to ETSI EN 302 636-6-1, except where explicitly stated otherwise.

Support column

The support column shall be filled in by the supplier of the implementation. The following common notations, defined in ISO/IEC 9646-7, are used for the support column:

Y or y	supported by the implementation.
N or n	not supported by the implementation.
N/A, n/a or -	no answer required (allowed only if the status is n/a, directly or after evaluation of a conditional status).

NOTE 2: As stated in ISO/IEC 9646-7, support for a received PDU requires the ability to parse all valid parameters of that PDU. Supporting a PDU while having no ability to parse a valid parameter is non-conformant. Support for a parameter on a PDU means that the semantics of that parameter are supported.

Values allowed column

The values allowed column contains the type, the list, the range, or the length of values allowed. The following notations are used:

- range of values: <min value> .. <max value>
  example: 5 .. 20

- list of values: <value1>, <value2>, ..., <valueN>
  example: 2, 4, 6, 8, 9
  example: '1101'B, '1011'B, '1111'B
  example: '0A'H, '34'H, '2F'H

- list of named values: <name1>(<val1>), <name2>(<val2>), ..., <nameN>(<valN>)
  example: reject(1), accept(2)

- length: size (<min size> .. <max size>)
  example: size (1 .. 8)

Values supported column

The values supported column shall be filled in by the supplier of the implementation. In this column, the values or the ranges of values supported by the implementation shall be indicated.

References to items

For each possible item answer (answer in the support column) within the ICS pro forma a unique reference exists, used, for example, in the conditional expressions. It is defined as the table identifier, followed by a solidus character "/", followed by the item number in the table. If there is more than one support column in a table, the columns are discriminated by letters (a, b, etc.), respectively.

EXAMPLE 1: A.5/4 is the reference to the answer of item 4 in table 5 of annex A.

EXAMPLE 2: A.6/3b is the reference to the second answer (i.e. in the second support column) of item 3 in table 6 of annex A.

Prerequisite line

A prerequisite line takes the form: Prerequisite: <predicate>.

A prerequisite line after a clause or table title indicates that the whole clause or the whole table is not required to be completed if the predicate is FALSE.

A.2.3 Instructions for completing the ICS pro forma

The supplier of the implementation shall complete the ICS pro forma in each of the spaces provided. In particular, an explicit answer shall be entered, in each of the support or supported column boxes provided, using the notation described in clause A.2.2.

If necessary, the supplier may provide additional comments in space at the bottom of the tables or separately.

More detailed instructions are given at the beginning of the different clauses of the ICS pro forma.

A.3 Identification of the implementation

A.3.1 Introduction

Identification of the Implementation Under Test (IUT) and the system in which it resides (the System Under Test (SUT)) should be filled in so as to provide as much detail as possible regarding version numbers and configuration options.

The product supplier information and client information should both be filled in if they are different.

A person who can answer queries regarding information supplied in the ICS should be named as the contact person.

A.3.2 Date of the statement

.....

A.3.3 Implementation Under Test (IUT) identification

IUT name:

.....

.....

IUT version:

.....

A.3.4 System Under Test (SUT) identification

SUT name:

.....

.....

Hardware configuration:

.....

.....

.....

Operating system:

.....

A.3.5 Product supplier

Name:

.....

Address:

.....

.....

.....

Telephone number:

.....

Facsimile number:

.....

E-mail address:

.....

Additional information:

.....

.....

.....

A.3.6 Client (if different from product supplier)

Name:

.....

Address:

.....

.....

.....

Telephone number:

.....

Facsimile number:

.....

E-mail address:

.....

Additional information:

.....

.....

A.3.7 ICS contact person

(A person to contact if there are any queries concerning the content of the ICS)

Name:

.....

Telephone number:

.....

Facsimile number:

.....

E-mail address:

.....

Additional information:

.....

.....

.....

A.4 Identification of the protocol

This ICS pro forma applies to the following standard: ETSI EN 302 636-6-1.

A.5 Global statement of conformance

Are all mandatory capabilities implemented? (Yes/No) .....

NOTE: Answering "No" to this question indicates non-conformance to the GEONET standard specification. Non-supported mandatory capabilities are to be identified in the ICS, with an explanation of why the implementation is non-conforming, on pages attached to the ICS pro forma.

A.6 Tables

A.6.1 Introduction

Unless stated otherwise, the column references of all tables below indicates the clause numbers of ETSI EN 302 636-6-1.

A.6.2 ITS Station type

A.6.2.1 General

Table A.1: ITS Station type

Item	Type	Reference	Status	Support
1	Central ITS station	6.1	o.101	
2	Road side ITS station	6.1	o.101	
3	Vehicle ITS station	6.1	o.101	
o.101:	It is mandatory to support one of these types.			

A.6.2.2 IPv6OverGeonetW sub layer

Table A.2: IPv6OverGeonetW sub layer

Item	Type	Reference	Status	Support
1	Is the IPv6OverGeonetW sub layer implemented and supported	6.1	c.201	
c.201:	IF A.1/3 THEN o ELSE m.			

A.6.2.3 Router type

Table A.3: Central ITS station

Prerequisite: A.1/1				
Item	**Type**	**Reference**	**Status**	**Support**
1	Application server	6.1	m	
2	Access router	6.1	x	
3	GeoAdhoc router	6.1	x	
4	Mobile router	6.1	x	

Table A.4: Road side ITS station

Prerequisite: A.1/2				
Item	**Type**	**Reference**	**Status**	**Support**
1	Application server	6.1	x	
2	Access router	6.1	o	
3	GeoAdhoc router	6.1	m	
4	Mobile router	6.1	x	

Table A.5: Vehicle ITS station

Prerequisite: A.1/3				
Item	**Type**	**Reference**	**Status**	**Support**
1	Application server	6.1	x	
2	Access router	6.1	x	
3	GeoAdhoc router	6.1	m	
4	Moblie router	6.1	o	

A.6.3 Virtual links

Table A.6: Virtual links

Item	**Type**	**Reference**	**Status**	**Support**
1	Static Geographical Virtual Link (SGVL)	5.2.1.1	Note 1	
2	Dynamic Geographical Virtual Link (DGVL)	5.2.1.2	Note 2	
3	Topological Virtual Link (TVL)	5.2.2	Note 3	
NOTE 1: It is mandatory to support at least one Static Geographical Virtual Link (SGVL). NOTE 2: It is mandatory to support exactly one Dynamic Geographical Virtual Link (DGVL). NOTE 3: It is mandatory to support exactly one Topological Virtual Link (TVL).				

A.6.4 Virtual network interfaces

Table A.7: virtual network interfaces

Item	**Type**	**Reference**	**Status**	**Support**
1	Ethernet V2.0/IEEE 802.3™ LAN	5.3.2.1	m	

A.6.5 Primitives Associated

Table A.8: primitives associated

Item	**Type**	**Reference**	**Status**	**Support**
1	GN6-UNITDATA.request	7	m	
2	GN6-UNITDATA.indication	7	m	

A.6.6 Packet Delivery

Table A.9: packet delivery

Item	**Type**	**Reference**	**Status**	**Support**
1	Outbound traffic	8.2.1	m	
2	Inbound traffic	8.2.2	m	

A.6.7 IPv6

A.6.7.1 IPv6 cast

Table A.10: IPv6 cast

Item	Type	Reference	Status	Support
1	IPv6 multicast	9.2	m	
2	IPv6 anycast	9.3	m	
3	Geographic IPv6 anycast	9.4	m	

A.6.7.2 IPv6 multicast

Table A.11: IPv6 multicast

Prerequisite: A.10/1				
Item	**Type**	**Reference**	**Status**	**Support**
1	IPv6 link-local multicast	9.2.1	m	
2	IPv6 wider-scope multicast	9.2.2	m	
3	Geocasting of IPv6 multicast traffic	9.2.3	m	

A.6.8 IPv6 neighbour discovery support

A.6.8.1 IPv6 neighbour discovery

Table A.12: IPv6 neighbour discovery

Item	Type	Reference	Status	Support
1	On-link determination	10.1	m	
2	Address configuration	10.2	m	
3	Address resolution	10.3	m	
4	Neighbour unreachability detection	10.4	m	

A.6.8.2 Address configuration

Table A.13: Address configuration

Prerequisite: A.12/2				
Item	**Type**	**Reference**	**Status**	**Support**
1	Stateless address auto configuration	10.2.1	m	
2	Stateful address configuration	10.2.2	m	
3	Manual address configuration	10.2.3	m	

A.6.8.3 Address resolution

Table A.14: Address resolution

Prerequisite: A.12/3				
Item	**Type**	**Reference**	**Status**	**Support**
1	Non-ND-based address resolution	10.3.1	m	
2	ND-based address resolution	10.3.2	m	

A.6.8.4 Constants

Table A.15: Constants

Item	Type	Reference	Status	Support
1	RTR_SOLICITATION_INTERVAL	10.5	m	
2	MAX_RTR_SOLICITATIONS	10.5	m	
3	MAX_MULTICAST_SOLICIT	10.5	m	
4	MAX_UNICAST_SOLICIT	10.5	m	
5	REACHABLE_TIME	10.5	m	

A.6.9 Protocol parameters

Table A.16: Protocol parameters

Item	Parameters	Ref.	Stat.	Sup.	Value allowed	Value supported
1	itsgn6aslVLTable	Annex A	m		itsgn6aslVLEntry (0..n)	
2	itsgn6aslTSversion	Annex A	m		ETSI EN 302 636-6-1 (1.2.x)	

Table A.17: itsgn6aslVLEntry

Item	Parameters	Ref.	Stat.	Sup.	Value allowed	Value supported
1	itsgn6aslVLType	Annex A	m		INTEGER	
2	itsgn6aslAreaPos1Latitude	Annex A	m		Integer32	
3	itsgn6aslAreaPos1Longitude	Annex A	m		Integer32	
4	itsgn6aslAreaPos2Latitude	Annex A	m		Integer32	
5	itsgn6aslAreaPos2Longitude	Annex A	m		Integer32	
6	itsgn6aslAreaDistA	Annex A	m		Unsigned32 (1..4 294 967 295)	
7	itsgn6aslAreaDistB	Annex A	m		Unsigned32 (1..4 294 967 295)	
8	itsgn6aslAreaAngle	Annex A	m		INTEGER (0..255)	
9	itsgn6aslVIIndex	Annex A	m		Ipv6IfIndex	
10	itsgn6aslVIResolAddr	Annex A	m		Truth Value	

History

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